Introduction

Every applied statistics course opens with a distinction that sounds obvious but is routinely blurred in papers: **descriptive** statistics summarise the sample at hand; **inferential** statistics make claims about the population from which the sample was drawn. Confusing them is the single most common reporting error in published research.

Prerequisites

The reader should know what a sample and a population are, and have encountered at least one summary statistic (mean, proportion).

Theory

A **descriptive statistic** is a numerical summary computed from the sample: mean, SD, median, IQR, counts, percentages. It is deterministic given the data; the same dataset always produces the same descriptive summary.

An **inferential statistic** uses the sample to say something about the population: a confidence interval, a hypothesis test, a regression coefficient with a standard error, a Bayesian credible interval. It carries an explicit or implicit uncertainty quantification.

The distinction maps cleanly onto notation: descriptive statistics (Latin letters $\bar{x}$, s, $\hat{p}$) versus population parameters (Greek letters μ, σ, π) that inferential procedures estimate.

Both play complementary roles in a paper:

- **Descriptive**: Table 1 characterising the sample; figures showing distributions.
- **Inferential**: Primary analysis results; p-values and confidence intervals for effect sizes.

Assumptions

Descriptive statistics require only that the data be correctly transcribed. Inferential statistics require a sampling-model assumption: that the sample reasonably represents the population (random sampling, randomised assignment, or a defensible non-random scheme).

R Implementation

```r
library(broom)
set.seed(2026)

trial <- data.frame(
  arm    = factor(rep(c("Placebo", "Active"), each = 60)),
  sbp    = c(rnorm(60, 138, 14), rnorm(60, 130, 14))
)

# Descriptive: summarise the sample
aggregate(sbp ~ arm, data = trial,
          FUN = function(x) c(n = length(x), mean = mean(x), sd = sd(x)))

# Inferential: make a claim about the population
fit <- t.test(sbp ~ arm, data = trial)
tidy(fit)
```

Output & Results

```
      arm n    mean    sd
```

```
 Placebo 60  137.82 14.05
  Active 60  130.39 13.28

# A tibble: 1 x 10
  estimate estimate1 estimate2 statistic p.value parameter conf.low conf.high
     <dbl>     <dbl>     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl>
1     7.43      138.      130.      2.99 0.00333      118.     2.52      12.3
```

Descriptive: placebo arm had a mean SBP of 137.8 mmHg (SD 14.1); active arm 130.4 (SD 13.3). Inferential: the difference is 7.4 mmHg with a 95 % CI of 2.5 to 12.3 mmHg, $p = 0.003$.

Interpretation

A paper's **Results** section typically progresses from descriptive to inferential: first describe the sample (Table 1), then test the pre-specified hypotheses. Mislabelling the two – treating a confidence interval as if it described the sample, or treating a sample mean as a population parameter – is a common reviewer complaint.

Practical Tips

- When you write "mean SBP was 137.8 mmHg" you have made a descriptive statement; when you write "SBP is 137.8 mmHg in the population" you have made an inferential statement (and should include a CI).
- Inferential statistics depend on a well-defined sampling frame; when the sample is a convenience sample, the inference is about the sample itself, not a broader population.
- Use distinct notation in equations: $\bar{x}$ for the sample mean, μ for the population mean.
- Every p-value and every confidence interval should be accompanied by a descriptive summary (counts, means, proportions) so the reader has both the effect and its context.
- A purely descriptive paper (audit, registry summary) is legitimate; pretending it is inferential is not.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1